HASSAM ASLAM

Unreal Engine Developer · C++ / Blueprints

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SUMMARY

Unreal Engine developer working in UE 5.8 C++ and Blueprints: combat, the Gameplay Ability System, enemy and NPC AI, multiplayer and UI. Shipped 12 Fortnite islands as Zeritress (855M+ minutes played). I build systems designers can tune through data, and I prove they work with automation tests and measured results.

SKILLS

Engine & code	Unreal Engine 5 (to 5.8), C++, Blueprints, Verse (UEFN), Visual Studio, Git
Gameplay	Gameplay Ability System (abilities, effects, attribute sets, tags, cues, prediction), combat with blocks and parries, input buffering, cameras, Enhanced Input, inventory, crafting and trading
AI	AI Controllers, behavior trees with custom C++ nodes, EQS, AI Perception, NavMesh and Detour crowd, NPC daily routines, LLM integration and retrieval
Multiplayer	Server-authoritative replication, client prediction, lag compensation, custom relevancy, owner-only and fast-array replication, server-side validation and rate limiting, listen and dedicated servers
UI, tools & testing	UMG written in C++, editor commandlets and Python editor scripts that generate assets, scripted multi-client network tests, automation tests, debug overlays and console commands

EXPERIENCE

UEFN Developer · Zeritress, freelance with creator partners — 2023 – present

- Shipped **12 Fortnite islands** with 6 creator partners: **855M+ minutes played**, a 44,962 all-time player peak, 482 published versions.
- Built Verse systems: persistent economies and save state, shops and item granting, per-player state tested with 50 players, prestige and ranked progression.
- Ran live-ops at about two updates a week on flagship islands, including PlayStation vs Xbox (552M minutes played).

AI Programmer & Technical Animator · Moshpit Studios

- NPC behaviour for a multiplayer virtual concert with AI Controllers and behavior trees; technical animation for crowd and player characters; tested NPCs against player powers.

C++ Contributor · Malakhra: Blood & Mayhem, 6-person team, UE 5.8 — 2026

- Built menu-hub systems in C++ (fighter cages, vanity wall, options, clickable navigation) and an editor-only camera-anchor visualizer for level artists.

SELECTED PROJECTS

Field Correspondent · solo, UE 5.8 C++ — 2026

- LLM-driven NPC: the model proposes actions through tool calls, C++ checks them against a whitelist and the NavMesh, and a behavior tree with six custom C++ nodes carries them out. Interrupts in about 0.2 s.
- Hybrid retrieval with citations (0.044 ms median search), AI Perception into timed memories. **120 automation tests** and 50 end-to-end checks passing in a packaged build.

Ashford Hollow · solo, UE 5.8 C++, 2–4 player networked RPG — 2026

- About **49,000 lines of C++**, every map and behavior tree generated by editor commandlets: GAS combat with statuses, blocks and parries, a two-phase boss, five streamed zones, quests, items, crafting and player trading saved per player.
- AI: one AI Controller with Perception, five behavior trees with six custom C++ nodes, EQS positioning, a wolf pack with attack tokens, ambushers, and villagers whose daily routines come from a schedule service and one replicated struct each.
- Netcode measured with scripted test clients: server-side rewind **cut ghost hits from 30% to 2% at 300 ms**; timestamped guard presses and a server hold took last-moment parries at 100 ms from **0 of 22 to 34 of 37**.
- Parties replicated to their members only, and chat the server rate-limits, cleans and filters line by line.

GASShooter port to UE 5.8 — 2026

- Ported tranek's GAS shooter sample from 5.3 to 5.8 (removed montage replication fields, instancing policy changes, prediction keys) and fixed an inverted ensure that blocked server-side target-data callbacks.

Third-person action RPG · UE5, carried forward in UE 5.8 C++ — 2026

- Melee combat with elemental abilities, enemy AI, XP and a three-branch skill tree across two levels; in UE 5.8, a soulslike Stone Golem boss fight and a new player character retargeted at runtime onto the existing combat.
- Inventory, crafting and loot in C++ wired to Blueprints by reflection; a Dark Souls-style C++ HUD; **10 automation tests**.